



SMART AUTONOMOUS HOSPITAL DELIVERY & SAFETY MONITOR ROBOT

CSE461: Introduction to Robotics | Section 8 | Group 3 | Summer 2026

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GROUP MEMBERS

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1. ABSTRACT

An autonomous hospital robot that delivers medicines and monitors safety in hospital environments. It uses a 5-channel IR array for line following, an underside MFRCS22 reader for RFID-based junction and destination identification, an HC-SR04 ultrasonic sensor for obstacle stopping, both flame and smoke hazard monitoring, a servo lock for the medicine box, and a 16x2 LCD with push-button destination selection.



2. INTRODUCTION

- Background:** Hospitals need fast, contactless delivery of medicines and real-time safety monitoring.
- Problem:** Manual delivery is time-consuming and labor-intensive.
- Objectives:**
 - Autonomous line-following navigation
 - RFID-based junction & destination ID
 - Obstacle detection and safe stop
 - Flame & smoke hazard monitoring
 - Secure medicine box with servo lock
 - Status display on 16x2 LCD
- Relevance:** Supports efficient logistics and enhances safety in hospital environments.



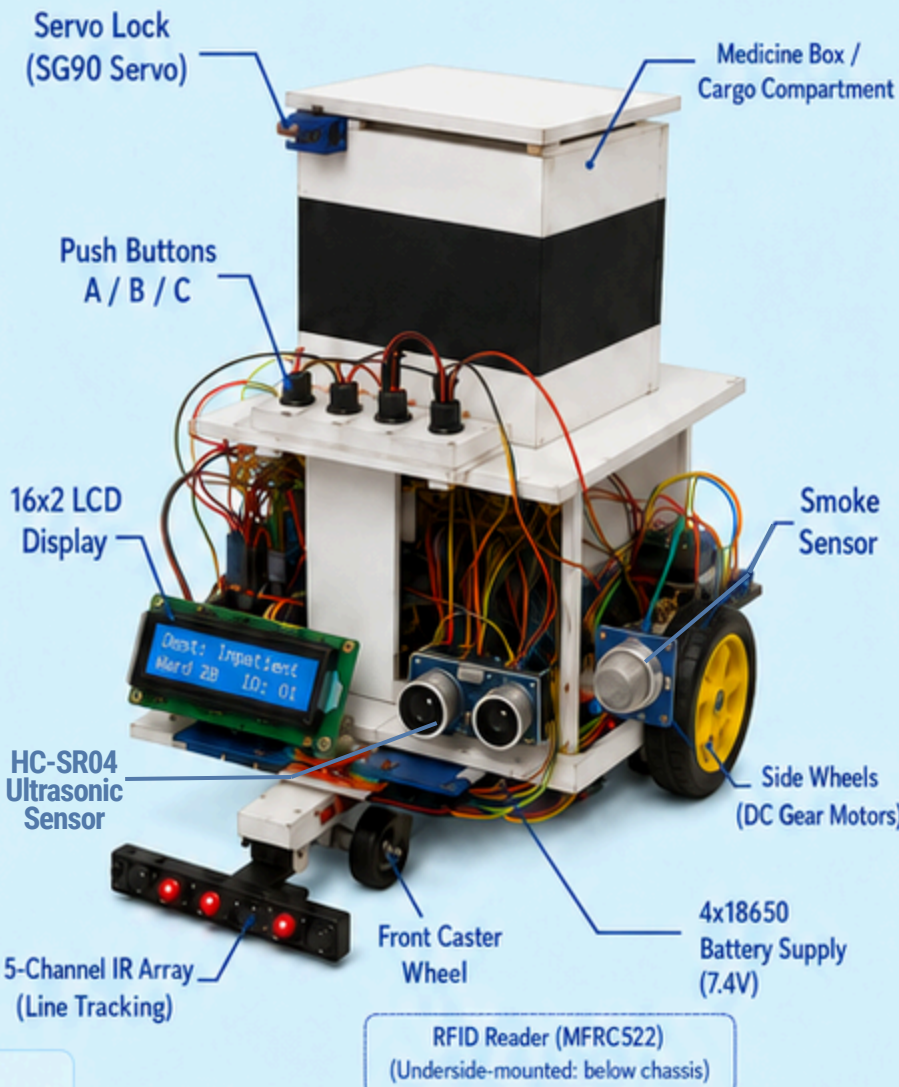
4. IMPLEMENTATION

CONTROL CIRCUIT DIAGRAM



HARDWARE COMPONENTS

- Arduino Uno R3
- L298N Motor Driver
- MFRCS22 RFID Reader
- HC-SR04 Ultrasonic Sensor
- 5-Channel IR Array
- SG90 Servo Motor (Lock)
- Smoke Sensor (MQ-2)
- IR Flame Sensor
- 16x2 LCD Display (I2C)
- PCF8574 I/O Expander
- 5V Active Buzzer
- Push Buttons
- 4x 18650 Li-ion cells / 7.4V supply



KEY FEATURES

- ✓ Follows black line tracks in hospital corridors
- ✓ RFID for junction & destination identification
- ✓ HC-SR04 obstacle detection & safe halt
- ✓ Servo lock secures medicine box
- ✓ 16x2 LCD displays real-time status
- ✓ Flame & smoke detection for early hazard warning

6. WORKFLOW / DECISION FLOW



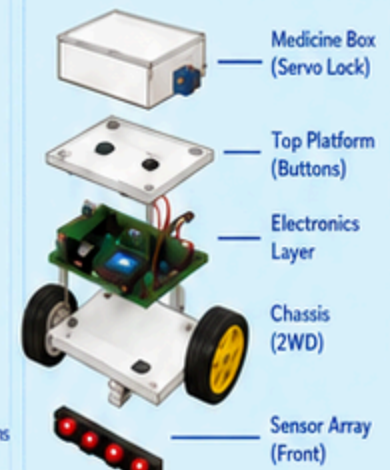
3. SYSTEM OVERVIEW

CONCEPT



- Robot follows line track
- RFID tags at junctions & destinations
- Obstacle detected -> Stop safely
- Deliver & continue

STRUCTURAL VIEW



5. FUNCTIONALITIES

MECHANICAL

- 2WD chassis with caster wheel
- Medicine box with servo lock
- Push buttons for manual control



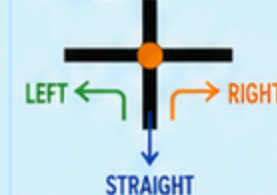
SOFTWARE

- Line following using 5-Channel IR
- RFID-based navigation
- Obstacle detection logic
- LCD status display
- Decision: Left / Straight / Right at junctions



NAVIGATION

- RFID tags at junctions & destinations
- Robot decides path: Left / Straight / Right
- Stops at destination and waits



SOFTWARE-HARDWARE COMMUNICATION

- I2C: LCD + PCF8574
- SPI: MFRCS22 RFID
- Digital/PWM I/O: motors, servo, ultrasonic, IR, smoke, flame
- Sensor data -> Arduino -> actuators / display



7. APPLICATIONS & COMPARISON

APPLICATIONS	ALTERNATIVES / COMPETITORS	ADVANTAGES	LIMITATIONS
- Medicine delivery between pharmacy & wards - Lab sample / reports transport - Food & supply delivery - Safety monitoring in restricted areas	- Manual Cart - AGVs (Industrial grade) - Commercial Delivery Robots Cost High Medium Low Low (Our Robot)	- Low cost & easy to build - Line + RFID hybrid navigation - Obstacle detection for safety - Flame & smoke monitoring - Real-time status display - Modular & upgradable	- Works on predefined line tracks - Indoor environment only - Limited payload capacity - Battery runtime limited - RFID tag deployment needed



8. CONCLUSION

SUMMARY
The robot successfully integrates line tracking, RFID navigation, obstacle detection, servo locking, and flame & smoke hazard monitoring in a compact and low-cost platform suitable for hospital environments.

KEY ACHIEVEMENTS

- ✓ Stable line following with 5-Channel IR
- ✓ Accurate RFID-based navigation
- ✓ Safe stop with HC-SR04
- ✓ Secure medicine box with servo lock
- ✓ Flame & smoke hazard monitoring
- ✓ Clear status on 16x2 LCD



FUTURE IMPROVEMENTS

- Add / re-integrate DHT11 temperature and humidity sensing
- Implement full RGB status indication
- Wireless monitoring (Wi-Fi / Bluetooth)
- More destinations & dynamic routing
- Auto-return to charging station
- Better chassis & higher payload



Smart Logistics for Safer Hospital Operations.

